



Global Cybersecurity Threats

BY

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Introduction

Cyberattacks are increasing and affecting organizations across different countries. Different attack types can cause financial loss and affect a large number of users. Some attacks also require more time to recover. Understanding attack patterns can help organizations improve their security. This project uses Power BI to analyze global cybersecurity threat data.

Main Focus

- ▶ Attack trends
- ▶ Financial losses
- ▶ Affected users
- ▶ Recovery time
- ▶ Industries and countries
- ▶ Defence methods

Data Cleaning

Tools:

- ▶ Power bi - data modelling, analysis and dashboard visualization.
- ▶ Power Query – data cleaning and transformation;
- ▶ MySQL – data storage and querying

Data Cleaning:

- ▶ Load data into Power Query.
- ▶ Add required columns.
- ▶ Create dimension tables from the fact table.
- ▶ Model the tables using relationships.
- ▶ Prepare the data for Power BI analysis.

Problem Statement & Objectives

Problem Statement

The main aim is to identify cyberattacks with long recovery times, strengthen defences to reduce attack frequency, and minimize financial losses.

Objectives

- ▶ Identify the most common attack types.
- ▶ Analyze yearly attack and financial-loss trends.
- ▶ Identify industries with high financial losses.
- ▶ Analyze countries with high attack impact.
- ▶ Study affected users.
- ▶ Compare attack sources and recovery time.
- ▶ Understand commonly used defence methods.
- ▶ Identify vulnerabilities frequently exploited by attacks.

Dataset

The project uses a cybersecurity threat dataset containing information related to:

- ▶ Attack ID
- ▶ Country
- ▶ Year
- ▶ Attack Type
- ▶ Target Industry
- ▶ Attack Source
- ▶ Financial Loss
- ▶ Affected Users
- ▶ Recovery Time
- ▶ Security Vulnerability Type
- ▶ Defence Mechanism

Star Schema

Fact Table

- ▶ Attack ID
- ▶ Country ID
- ▶ Measures such as financial loss, affected users and recovery time

Dimension Tables

- ▶ Country Dimension
- ▶ Attack Dimension

Purpose

The star schema makes the data easier to filter, analyze and visualize in Power BI

Dashboard Overview

The dashboard provides a summary of the cybersecurity dataset through KPIs and charts.

Main KPIs

- ▶ Total Attacks: 100K
- ▶ Average Affected Users: 504.67K
- ▶ Average Recovery Time: 36.48 hours
- ▶ Peak Attack Year: 2017
- ▶ Total Financial Loss: shown in the dashboard

The overview page also analyzes:

- ▶ Yearly financial loss
- ▶ Attack types over the years
- ▶ Affected users by year
- ▶ Target industries by year
- ▶ 2023–2024 comparison

Yearly Trend Analysis

The dashboard uses yearly charts to understand how cyber threats change over time.

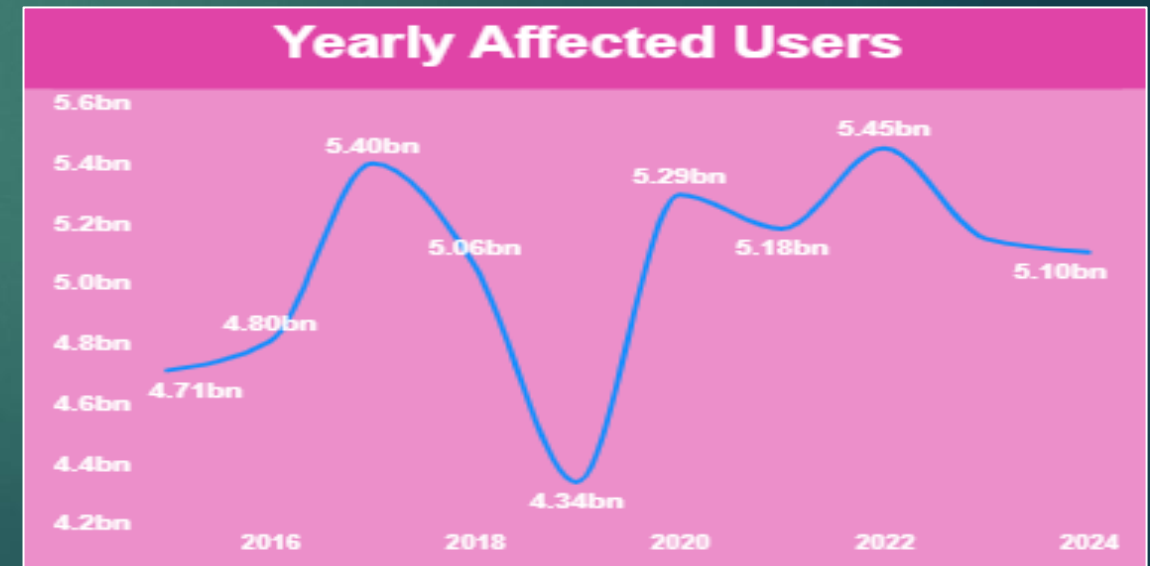
Key Analysis:

Yearly Financial Loss

- ▶ Financial loss changes from year to year.
- ▶ The dashboard shows noticeable increases and decreases across the period.

Affected Users by Year

- ▶ The number of affected users varies significantly between years.
- ▶ Some years show a much higher impact than others.



Attack Types

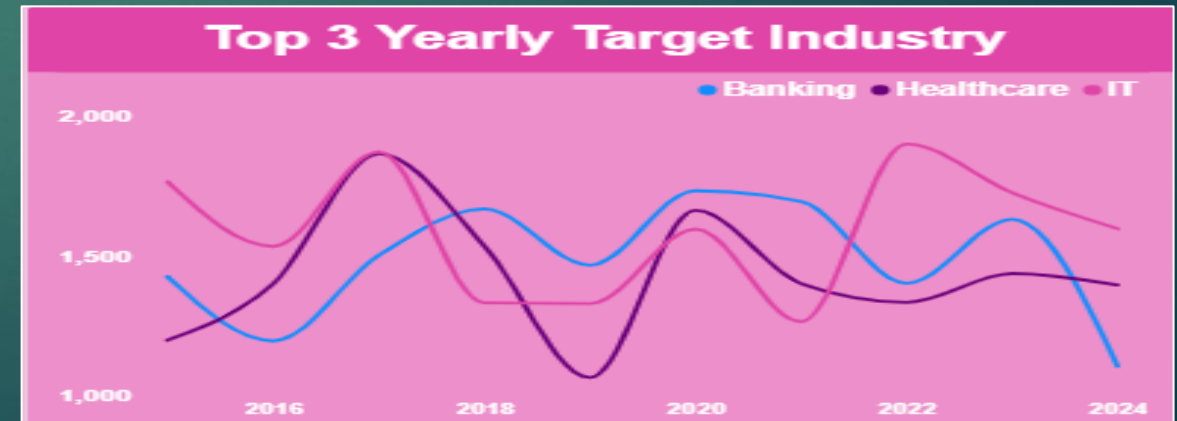
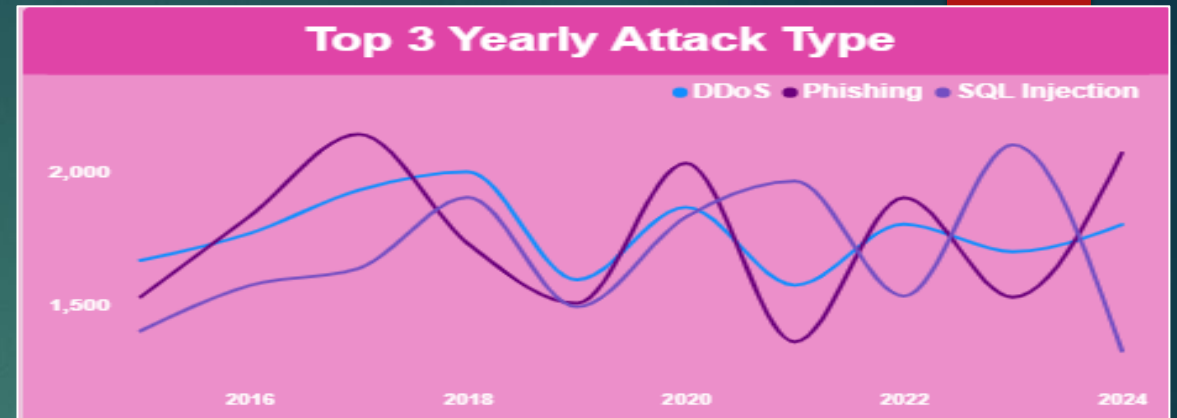
- Different attack types show different patterns over the years.
- DDoS, Phishing and SQL Injection are among the attack types displayed.

Peak Attack Year

- 2017 is shown as the peak attack year

Target industry

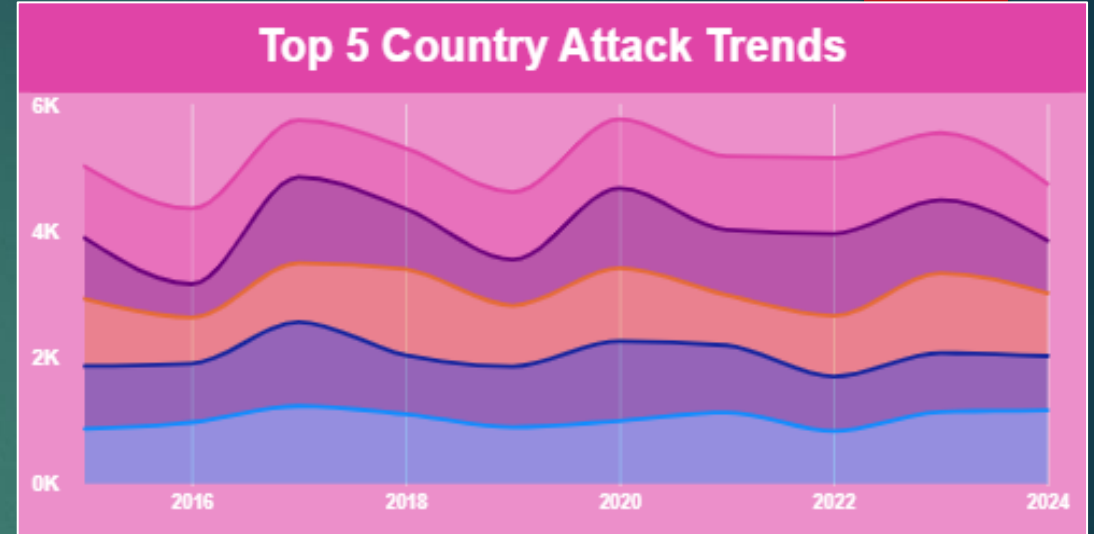
- The Top 3 Yearly Target Industry chart shows how attacks against Banking, Healthcare and IT change over time.
- It helps identify which industries experience higher attack activity in different years.



COUNTRY & USER ANALYSIS

Top 5 Country Attack Trends

- The Top 5 Country Attack Trends chart compares attack activity across the leading countries over time.
- The area size shows the relative number of attacks for each country.



Financial Impact

- The Financial Impact map gives a geographical view of cyberattack losses.
- It helps us understand which countries have greater financial impact.



Users Impact by Country

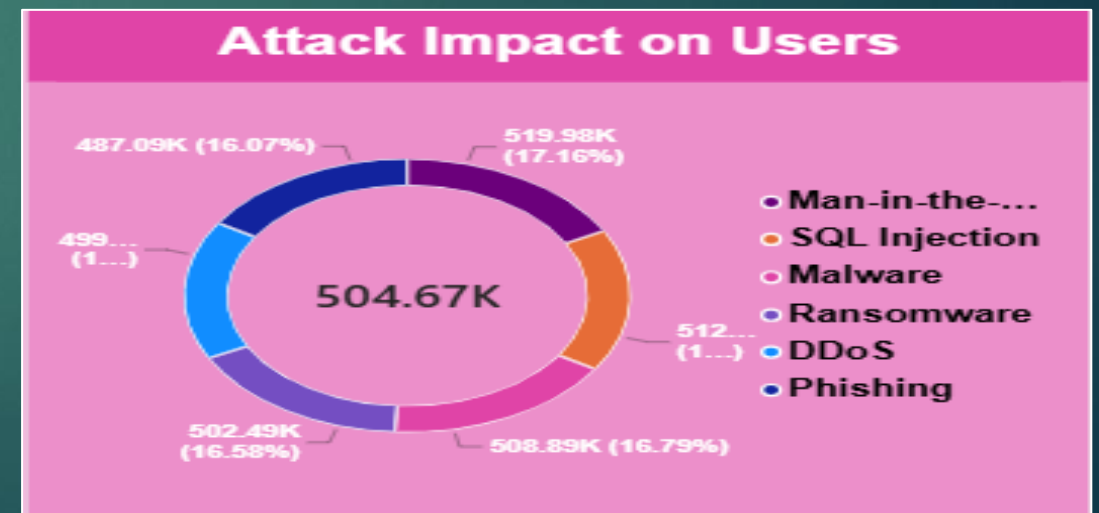
- The Users Impact by Country map shows how many users are affected in different countries.
- According to the dashboard, Brazil has the highest user impact.

Attack Source by Country

- The Attack Source by Country map shows the geographical distribution of different attack sources.
- It helps us understand the relationship between countries and sources of cyberattacks.

Attack Impact on Users

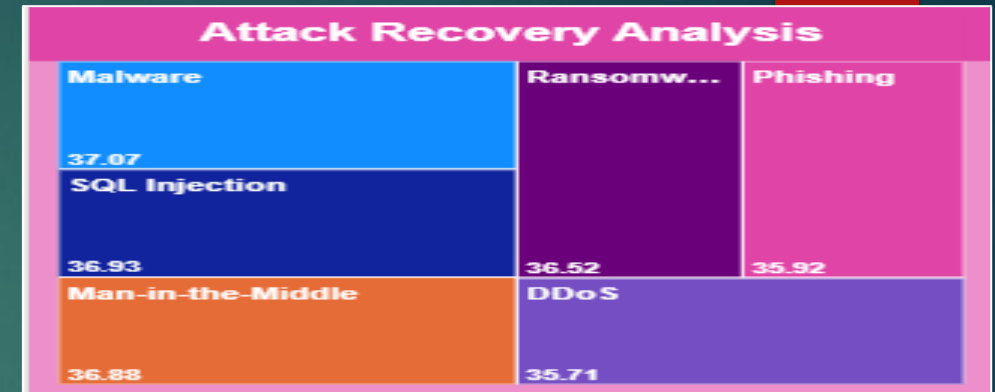
- The Attack Impact on Users donut chart compares how different attack types affect users.
- It helps identify which types of attacks are associated with higher user impact.



ATTACK & INDUSTRY ANALYSIS

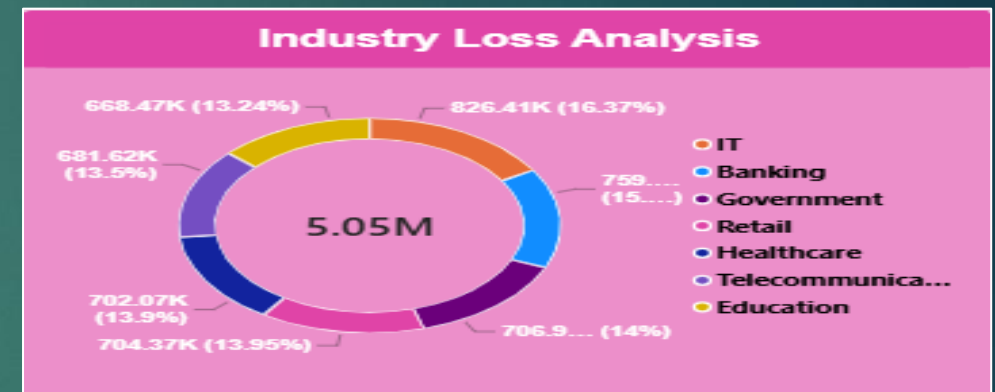
Attack Recovery Analysis

- The Attack Recovery Analysis compares the average recovery time for different attack types.
- Malware has the highest displayed recovery time, while DDoS has the lowest among the attack types shown.



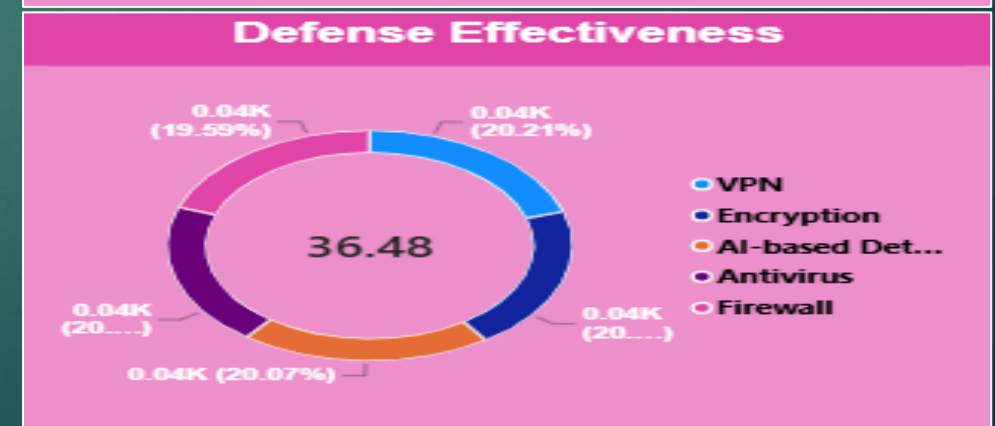
Industry Loss Analysis

- The Industry Loss Analysis chart compares financial losses across different industries.
- IT is identified as the top loss industry.



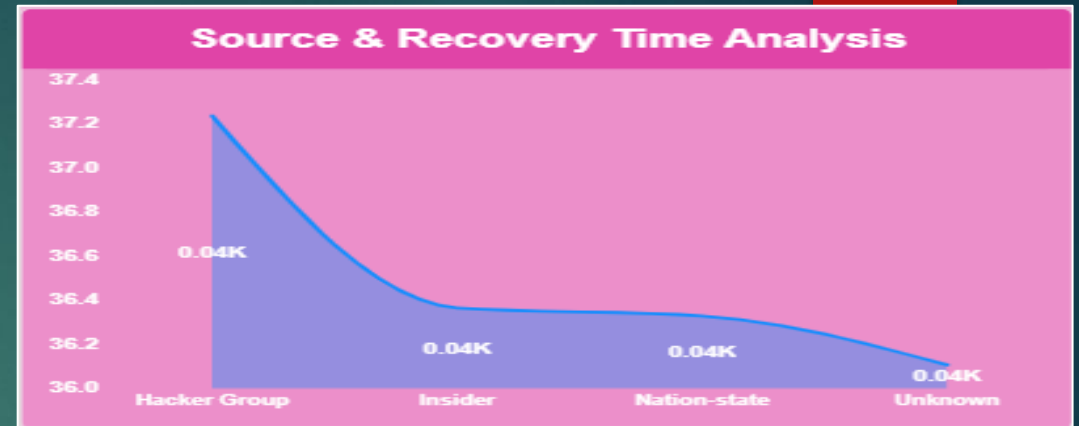
Defense Effectiveness

- The Defense Effectiveness chart compares different defense mechanisms used against cyberattacks.
- The Chart identifies VPN as the top defense method.



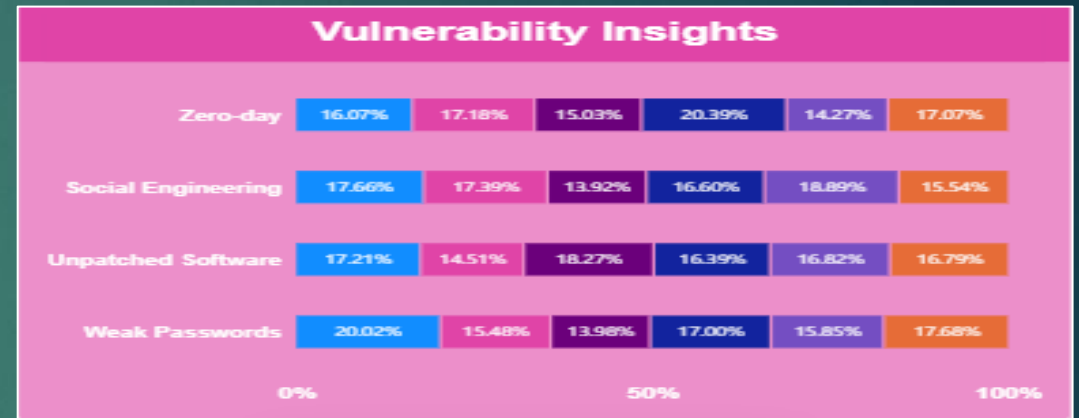
Source & Recovery Time Analysis

- The Source and Recovery Time Analysis compares recovery time across different attack sources.
- It helps identify which sources are associated with higher recovery-time values.



Vulnerability Insights

- The Vulnerability Insights chart compares different security vulnerabilities.
- It helps us understand how vulnerability categories are distributed across the attack data.
- Unpatched Software is highlighted as the top vulnerability in the dashboard.



Source & Attack

- The Source and Attack chart shows the relationship between attack sources and attack activity.
- It helps us understand how different sources contribute to cybersecurity incidents.



Recommendations & Future Scope

Based on the dashboard analysis:

- ▶ Regularly update and patch software.
- ▶ Monitor high-risk attack types such as DDoS.
- ▶ Strengthen network security.
- ▶ Improve employee security awareness.
- ▶ Monitor suspicious attack sources.
- ▶ Use appropriate defense mechanisms.
- ▶ Continuously monitor financial and user impact.

Future Scope

- ▶ Add real-time cybersecurity data.
- ▶ Include more countries and industries.
- ▶ Use machine learning for attack prediction.
- ▶ Create automated threat alerts.
- ▶ Analyze emerging attack types.
- ▶ Develop a real-time cybersecurity monitoring dashboard.

Conclusion

- ▶ The project provides a complete view of global cybersecurity threats.
- ▶ Power BI helps convert cybersecurity data into meaningful visual insights.
- ▶ The dashboard identifies important patterns in attacks, financial loss, users, industries, countries and defense methods.
- ▶ The analysis can help organizations understand their major cybersecurity risks.
- ▶ The insights can support better security planning, monitoring and risk reduction.

Thank you